

IRREGULAR CLUSTER

Hercules Cluster



CATALOG NUMBER
Abell 2151

DISTANCE
500 million light-years

NUMBER OF GALAXIES
100+

BRIGHTEST MEMBER
NGC 6041A (14.4)

HERCULES

The small Hercules Cluster is dominated by spiral and irregular galaxies, suggesting that it is in an early

stage of development. In keeping with the best models of such clusters' formation (see p.327), it shows little sign of structure. Within the cluster, several pairs or groups of galaxies seem to be merging or interacting—encounters that will transform them into different kinds of galaxies and reduce their random movements until they become more evenly distributed. The most prominent of these mergers is NGC 6050, a pair of interlocking spiral galaxies near the cluster's center that may eventually form the core of a giant elliptical, such as those found in more evolved clusters.

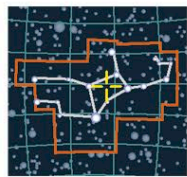
HERCULES FIELD

This wide-field view captures most of the bright galaxies in Hercules and shows their irregular, "unrelaxed" distribution.



REGULAR CLUSTER

Abell 1689



CATALOG NUMBER
Abell 1689

DISTANCE
2.2 billion light-years

NUMBER OF GALAXIES
3,000+

BRIGHTEST MEMBER
2MASX J13112952-0120280 (16.5)

VIRGO

Abell 1689 is one of the densest galaxy clusters known, with thousands of galaxies packed into a volume of space only 2 million light-years across. Its ball shape makes it a fine gravitational lens. By measuring the lensing power throughout the cluster, astronomers have worked out the distribution of the cluster's dark matter.



DARK MATTER MAPPED IN ABELL 1689

GEORGE ABELL

George Abell (1927–1983) was a career astronomer and popularizer of science who carried out the first and most influential survey of galaxy clusters. After working on the Palomar Sky Survey during the 1940s and 1950s, using the Palomar Schmidt telescope, he turned his attention to analyzing the results, developing methods for distinguishing galaxy clusters from isolated field galaxies and classifying clusters into types.



REGULAR CLUSTER

Abell 2065



CATALOG NUMBER
Abell 2065

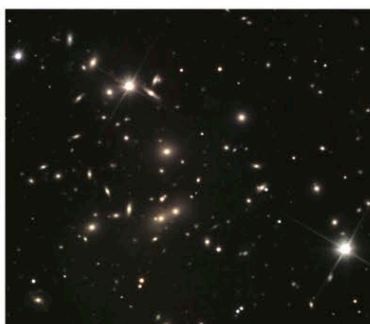
DISTANCE
1 billion light-years

NUMBER OF GALAXIES
1,000+

BRIGHTEST MEMBER
PGC 54876 (16.0)

CORONA BOREALIS

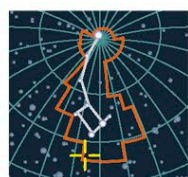
Abell 2065, also known as the Corona Borealis Cluster, contains 400 or more large galaxies. A highly evolved cluster like the Coma Cluster (opposite), it emits X-rays from a diffuse cloud of hot gas. However, X-ray observations have found two distinct X-ray cores, suggesting that Abell 2065 may be two already ancient clusters merging together. The cluster lies at the center of the Corona Borealis Supercluster.



THE CORONA BOREALIS CLUSTER

IRREGULAR CLUSTER

Abell 2125



CATALOG NUMBER
Abell 2125

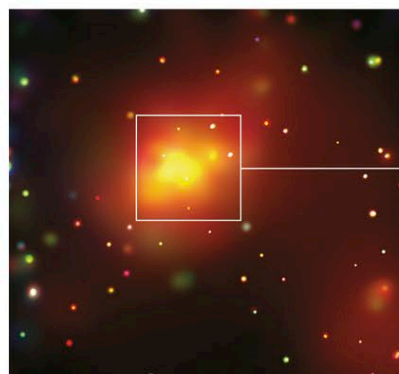
DISTANCE
3 billion light-years

NUMBER OF GALAXIES
1,000+

BRIGHTEST MEMBER
Magnitude 17.0

URSA MINOR

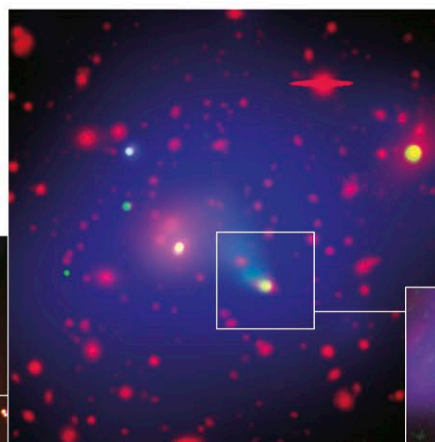
Abell 2125 has been the subject of intense scrutiny from the orbiting Chandra X-Ray Observatory. The cluster lies close enough to Earth to see detail but so far away that images reaching Earth show an early and still active phase of its evolution, 3 billion years ago. Abell 2125 is therefore ideal for testing ideas on cluster formation.



X-ray images reveal what optical ones cannot—that the cluster is forming from the merger of several smaller clusters. The most intense cloud of X-ray-emitting gas shows "clumpiness," which indicates it has recently come together. Spectra reveal that the cloud is enriched with heavy elements such as iron, and close-up images show gas actively being stripped away from galaxies such as C153. With it, the gas carries atoms of heavy metals created in supernova explosions, distributing

them through the intergalactic medium. A fainter cloud of almost equal size, enveloping hundreds more galaxies, has remarkably few heavy elements, suggesting that the gas-stripping process becomes more powerful and thorough over time and that the cloud is much younger than its fainter neighbor.

Because X-ray evidence shows so much activity within the cluster, astronomers have also imaged it at other wavelengths. Infrared telescopes, for example, have revealed enormous bursts of star formation going on in galaxies far from the cluster center. One possible explanation is that, even at distances of up to 1 million light-years, the tidal forces from the center of a large cluster are enough to disrupt nearby galaxies and trigger starbursts.



ZOOMING IN ON C153

This sequence of Chandra X-ray images zooms into the hot gas cloud at the core of Abell 2125, showing how gas is being stripped from galaxy C153 (right).

